

## NEW YORK KENNEDY, NY, USA

WMO: 744860

Lat: 40.639N

Lon: 73.764W

Elev: 11

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94789

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 13.5       | 17.5 | -6.0                          | 4.0 | 16.3 | -1.8 | 5.0 | 20.9 | 32.6                  | 29.2 | 29.4 | 30.2 | 16.5                  | 300  | 0.573 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.4                   | 89.8            | 73.0 | 86.7 | 72.1 | 84.2 | 71.2 | 77.3                | 83.7 | 75.9 | 81.9 | 74.7 | 80.4 | 13.0                 | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 75.4                            | 133.3 | 80.3 | 74.1 | 127.3 | 79.1 | 72.8 | 121.6 | 78.0 | 40.6          | 83.4 | 39.2 | 82.1 | 38.1 | 80.5 | 84.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |     |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|-----|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   | DB  | WB  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |     |     |
| (4) 27.8          | 25.0 | 22.5 | 8.2                        | 95.8 | 5.0                | 3.4 | 4.6                                                | 98.2 | 1.7        | 100.3 | -1.1       | 102.2 | -4.8       | 104.7 |     |     |
| (5)               |      |      | 5.8                        | 79.5 | 4.4                | 2.1 | 2.6                                                | 81.1 | 0.0        | 82.3  | -2.5       | 83.5  | -5.7       | 85.0  | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 54.9   | 33.0  | 34.8  | 41.5  | 51.5  | 60.7  | 70.5  | 76.4  | 75.3  | 69.1  | 58.0  | 47.4  | 38.9  |  |  |
|                                                                        | DBStd       | 16.65  | 9.37  | 8.23  | 7.77  | 6.67  | 6.22  | 5.97  | 4.65  | 4.49  | 5.58  | 6.97  | 7.49  | 8.12  |  |  |
|                                                                        | HDD50       | 1801   | 530   | 427   | 280   | 58    | 1     | 0     | 0     | 0     | 0     | 13    | 136   | 356   |  |  |
|                                                                        | HDD65       | 4761   | 994   | 845   | 728   | 408   | 165   | 19    | 0     | 1     | 25    | 237   | 529   | 810   |  |  |
|                                                                        | CDD50       | 3573   | 2     | 3     | 17    | 102   | 333   | 614   | 817   | 786   | 572   | 260   | 57    | 10    |  |  |
|                                                                        | CDD65       | 1057   | 0     | 0     | 0     | 3     | 32    | 183   | 353   | 321   | 146   | 19    | 0     | 0     |  |  |
|                                                                        | CDH74       | 6799   | 0     | 0     | 3     | 32    | 179   | 1080  | 2670  | 2156  | 623   | 56    | 0     | 0     |  |  |
|                                                                        | CDH80       | 1833   | 0     | 0     | 0     | 7     | 35    | 302   | 826   | 550   | 102   | 11    | 0     | 0     |  |  |
|                                                                        | WSAvg       | 11.3   | 12.8  | 12.7  | 13.0  | 12.1  | 10.5  | 10.1  | 9.9   | 9.4   | 9.9   | 10.9  | 11.7  | 12.4  |  |  |
|                                                                        | PrecAvg     | 45.7   | 3.5   | 2.8   | 4.1   | 3.7   | 3.8   | 4.0   | 4.0   | 4.8   | 3.7   | 4.0   | 3.2   | 4.1   |  |  |
| (15) Precipitation                                                     | PrecMax     | 63.7   | 7.8   | 5.3   | 10.0  | 9.6   | 7.7   | 9.5   | 7.6   | 17.4  | 9.1   | 15.0  | 5.7   | 6.8   |  |  |
|                                                                        | PrecMin     | 35.9   | 1.4   | 0.7   | 0.9   | 1.4   | 1.5   | 0.5   | 0.8   | 0.6   | 1.2   | 0.4   | 0.9   | 1.0   |  |  |
|                                                                        | PrecStd     | 6.4    | 1.4   | 1.3   | 2.0   | 1.9   | 1.4   | 2.1   | 1.9   | 3.1   | 1.9   | 2.9   | 1.4   | 1.4   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 56.9  | 60.2  | 67.8  | 78.6  | 84.3  | 91.7  | 95.8  | 92.5  | 87.1  | 80.5  | 67.4  | 61.9  |  |  |
|                                                                        |             | MCWB   | 51.3  | 50.2  | 54.2  | 60.6  | 67.7  | 73.4  | 76.0  | 74.8  | 70.9  | 67.1  | 58.6  | 54.9  |  |  |
|                                                                        | 2%          | DB     | 52.3  | 53.4  | 60.6  | 70.4  | 78.9  | 87.0  | 90.8  | 88.3  | 82.9  | 74.0  | 63.8  | 57.0  |  |  |
|                                                                        |             | MCWB   | 49.3  | 46.0  | 49.4  | 55.7  | 65.1  | 71.4  | 73.4  | 73.1  | 69.8  | 65.2  | 57.8  | 52.9  |  |  |
|                                                                        | 5%          | DB     | 49.3  | 49.3  | 55.7  | 65.5  | 74.7  | 83.3  | 87.3  | 85.4  | 79.9  | 71.1  | 61.2  | 53.7  |  |  |
|                                                                        |             | MCWB   | 45.8  | 44.0  | 47.6  | 52.8  | 63.3  | 69.3  | 72.4  | 72.3  | 68.9  | 64.8  | 56.6  | 50.5  |  |  |
|                                                                        | 10%         | DB     | 45.9  | 46.1  | 52.0  | 61.6  | 70.9  | 80.0  | 84.3  | 82.9  | 77.4  | 68.3  | 58.7  | 50.8  |  |  |
|                                                                        |             | MCWB   | 42.2  | 41.5  | 45.2  | 51.3  | 61.0  | 68.0  | 71.8  | 71.4  | 68.5  | 62.1  | 54.2  | 46.9  |  |  |
|                                                                        | 0.4%        | WB     | 53.5  | 53.3  | 57.4  | 64.0  | 71.4  | 76.2  | 79.9  | 79.4  | 76.5  | 71.6  | 63.3  | 58.0  |  |  |
|                                                                        |             | MCDB   | 55.1  | 56.6  | 64.8  | 73.8  | 80.7  | 86.4  | 89.4  | 86.9  | 80.6  | 74.1  | 64.8  | 60.6  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 50.4  | 48.3  | 52.4  | 59.5  | 68.3  | 74.2  | 77.4  | 77.4  | 74.5  | 68.9  | 60.2  | 54.0  |  |  |
|                                                                        |             | MCDB   | 51.8  | 51.0  | 56.9  | 65.9  | 75.2  | 82.5  | 84.6  | 83.1  | 78.4  | 72.5  | 62.0  | 55.7  |  |  |
|                                                                        | 5%          | WB     | 46.4  | 45.1  | 49.4  | 56.4  | 65.8  | 72.4  | 76.1  | 76.0  | 72.9  | 66.4  | 57.8  | 51.3  |  |  |
|                                                                        |             | MCDB   | 48.6  | 48.0  | 54.1  | 61.5  | 71.9  | 79.2  | 82.6  | 81.2  | 76.5  | 69.9  | 60.1  | 53.6  |  |  |
|                                                                        | 10%         | WB     | 42.9  | 42.1  | 46.5  | 53.8  | 63.3  | 70.8  | 74.8  | 74.7  | 71.1  | 63.6  | 55.0  | 47.7  |  |  |
|                                                                        |             | MCDB   | 45.5  | 45.8  | 50.7  | 58.8  | 68.9  | 76.7  | 81.0  | 79.7  | 75.3  | 67.2  | 58.0  | 50.3  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.9  | 12.8  | 13.5  | 14.6  | 14.2  | 14.0  | 13.4  | 13.0  | 13.2  | 13.3  | 12.5  | 11.2  |  |  |
|                                                                        |             | MCDBR  | 15.0  | 16.4  | 18.6  | 20.8  | 19.5  | 18.7  | 16.6  | 15.3  | 14.8  | 15.2  | 14.4  | 14.1  |  |  |
|                                                                        |             | MCWBR  | 13.6  | 12.7  | 11.7  | 10.8  | 10.2  | 8.4   | 6.8   | 7.4   | 7.9   | 9.8   | 11.4  | 13.6  |  |  |
|                                                                        |             | MCDBR  | 14.5  | 15.7  | 16.7  | 17.3  | 17.2  | 16.0  | 13.6  | 12.6  | 11.7  | 13.3  | 12.6  | 13.6  |  |  |
|                                                                        | 5% WB       | MCWBR  | 14.4  | 14.1  | 12.7  | 11.4  | 10.3  | 8.5   | 6.6   | 6.9   | 7.8   | 10.5  | 12.2  | 14.5  |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.321 | 0.331 | 0.362 | 0.386 | 0.434 | 0.488 | 0.494 | 0.478 | 0.404 | 0.365 | 0.335 | 0.321 |  |  |
|                                                                        | taud        |        | 2.468 | 2.443 | 2.407 | 2.373 | 2.270 | 2.156 | 2.150 | 2.187 | 2.368 | 2.484 | 2.522 | 2.523 |  |  |
|                                                                        | Ebn at Noon |        | 266   | 280   | 282   | 281   | 269   | 254   | 251   | 251   | 265   | 264   | 259   | 257   |  |  |
|                                                                        | Edh at Noon |        | 25    | 29    | 34    | 37    | 42    | 48    | 47    | 44    | 35    | 28    | 23    | 22    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 593   | 880   | 1198  | 1539  | 1711  | 1880  | 1907  | 1683  | 1385  | 965   | 674   | 509   |  |  |
|                                                                        | RadStd      |        | 50    | 89    | 102   | 150   | 174   | 174   | 93    | 108   | 123   | 94    | 58    | 47    |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | +0.65 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +184  | +122  |
| (47) | Regional (63 neighbors) | +0.72 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +209  | +121  |

Nomenclature: See separate page